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Practical Session 1: Mutex Lock Demonstration

Objective
Demonstrate mutual exclusion using pthread mutex locks.

Answer / Explanation
1. Initialize a pthread_mutex_t variable.
2. Create multiple threads that increment a shared counter.
3. Protect the shared counter using pthread_mutex_lock() and pthread_mutex_unlock().
4. Compare the results with and without a mutex lock.
5. Observe that without a mutex, race conditions occur and the final counter value may be incorrect.

Sample C Program
```
#include <stdio.h>
#include <pthread.h>

int counter = 0;
pthread_mutex_t lock;

void* increment(void* arg) {
for(int i = 0; i < 100000; i++) {
pthread_mutex_lock(&lock);
counter++;
pthread_mutex_unlock(&lock);
}
return NULL;
}
```

Result:
With mutex lock: The final counter value is correct because only one thread accesses the shared resource at a time.

Without mutex lock: Race conditions occur, causing inconsistent and incorrect counter values.